# What is Democracy? Scalable Reading Newspapers of the Weimar Republic

**Wachter, Christian**
christian.wachter@uni-bielefeld.de
Bielefeld University, Germany

## Digital Newspaper Research and Cultural History of Politics of Interwar Germany

This paper introduces an ongoing project focused on analyzing newspaper discourses of the Weimar Republic (Wachter 2023a, 2023b). It features a workflow that combines quantitative digital analysis with textual hermeneutics, resulting in a scalable reading approach. This method responds to a critical need in historical research: Traditionally, historians have characterized interwar Germany's political development through a narrative of crisis. However, this perspective has increasingly been challenged by historians who highlighted the potential for stability and democratic culture. Despite this shift, prodemocratic attitudes within Weimar's primary opinion-forming medium, newspapers, remain underexplored due to methodological limitations. This is because traditional research methods struggle with the vast volume of newspapers, forcing historians to narrow their focus by region, period, or topic. In contrast, scalable reading enables historians to obtain more extensive, detailed, and nuanced insights into democratic thinking.

Scalable reading approaches have been applied to large volumes of newspapers in other historical contexts (e.g., Sung, Sun, and Arnold 2014; Keck, Knabben, and Pado 2020). For example, they concentrate on text reuse, which has also been highlighted as an important avenue for accessing cultural history (Salmi 2024). Other studies examine argument strategies in migration discourse (Oberbichler 2020, Oberbichler and Pfanzelter 2022), providing valuable inspiration for further newspaper discourse analysis. However, these approaches are inadequate for direct application to the Weimar case, where specialized scalable reading workflows for large newspaper volumes have never been applied (Giulia de Paduanis (2023) followed a similar approach in her master's thesis, but conducted only a small-scale analysis): The fragmented press landscape of interwar Germany was characterized by strong opinion journalism, featuring effective strategies of legitimation and delegitimation – also regarding ideas on democracy. In a scalable reading workflow, these strategies must be systematically identified as discourse linguistic characteristics, which has not been done before. Therefore, the ongoing project draws on Theo van Leeuwen's (2007) analytical differentiation of legitimation strategies that invoke (1) authority (e.g., the constitution), (2) moral evaluation (reinforcing ethical standards), (3) rationalization (defending institutions/measures as reasonable), and (4) mythopoesis (narratives rewarding legitimate actions and punishing non-legitimate ones). These discourse-analytic categories are central to the project's concept and methodological pipeline.

## Methodological Critique: Beyond the Text Surface

Terms such as "democracy" or "constitution" do not fully capture the underlying meanings and experiences that characterize discourse. Equally important is understanding what remains unsaid and why (Kutter 2017). In this context, searches for key terms and text mining must be supplemented by qualitative selections of relevant newspaper issues, such as those published before and after significant political events like elections. While distant reading is a critical exploratory element in the project, it alone does not suffice for this context sensitivity. Even quantitative analysis of absent terms and phrases proves insufficient within the framework of discourse analysis as conceptualized in this project, as it would merely operate on the text surface, ignoring the discursive findings that lie beneath it (cf. Kergomard 2022). For instance, in an article explicitly criticizing prewar German bureaucracy, we could identify an implicit endorsement of participatory politics for a democratized society.

## Pilot Studies

To test and refine the scalable reading workflow – not to engage in fully-fledged discourse research – the project will conduct three pilot studies: (1) social-democrat discourse on Constitution Day (the national holiday), (2) the city of Bielefeld's regional discourse on the murder of Foreign Minister Walter Rathenau, and (3) Berlin's discourse on the end of the First Portuguese Republic. For these pilot studies, digitized historical newspapers will be

accessed through portals such as zeit.punktNRW, digiPress, ZEFYS, the German Newspaper Portal, and Europeana Newspapers. While only a fraction of Weimar era newspapers has been digitized, the issues relevant to the pilot studies are widely available: 31,816 issues in total, minimizing the need to examine non-digitized archival material.

## OCR Cleaning

Historical newspapers often suffer from noisy OCR. Various approaches to OCR cleaning have been developed using smaller language models (e.g., Soper, Fujimoto, and Yu 2021; Rakshit, Mehta, and Dasgupta 2023). More recently, large language models have been introduced for this task, often outperforming smaller models with little or no need for finetuning (e.g., Oberbichler 2024; Thomas, Gaizauskas, and Lu 2024). Building on Oberbichler's (2024) work in this field, Christian Wachter and Patrick Jentsch from Bielefeld University's Digital History Working Group have developed a method for testing and employing various locally hosted LLMs, utilizing Ollama, to encance the OCR quality of the aforementioned newspaper issues with Python.[1]

## Corpus Compilation through Methodological Triangulation

The cleaned issues are to be imported into the annotation and analysis tool CATMA, where they undergo keyword searches. Relevant keywords are derived from research literature, and further terms are identified inductively in later analysis steps, entering new searches in an iterative workflow. Articles thus found become included in the corpus. Complementarily, topic modeling (LDA, with the tool MALLET) supports the exploration of additional relevant articles, as this method hints to potentially relevant word groups. These analysis results present democracy as a subject of discussion across various contexts – ranging from expectations towards the day-to-day work of parliamentarians, to the constitution as the political and ideological foundation of Weimar Germany, and the economy and administrative institutions as arenas of democratization. Like other distant reading methods, topic modeling is particularly effective at revealing unexpected findings – specifically, articles that were not published in connection with major politically significant events, where discourses of democracy would typically be expected. Furthermore, the LLM setup from the OCR-cleaning step also supports content analysis, uncovering discourse on democracy. This is achieved through few-shot prompting that seeks formulations which avoid explicitly mentioning "democracy" and related terms, instead operating through indirect statements (such as "we should do this…"). Additionally, van Leeuwen's archetypical (de-)legitimation strategies serve as analytical cornerstones for finding relevant press articles. Lastly, manual selections from newspaper portals and physical archives (digitized with the tool nopaque) will be added.

## Structuring the Corpus

The corpus is subjected to word frequency (relative/absolute) analyses in CATMA, which serve as starting points for close reading. This process aims to identify and annotate distinct discursive strategies and topoi in CATMA. The annotation scheme aligns with van Leeuwen's typology. CATMA's query function then allows the corpus texts to be accessed according to specific annotations, such as filtering for the annotation tag "praising the constitution" (reference to authority, according to van Leeuwen) or the tag "opposing the amalgamation of antidemocratic discourse and antisemitism" (moral evaluation). This process uncovers semantic connections between these passages. The free-to-use tool CATMA is particularly well-suited for these tasks. It offers flexible capabilities for searching the corpus, annotating texts, and visualizing and evaluating data within a browser-based environment. In addition, CATMA supports exporting data for further analysis with external tools.

## Source Critique and Interpretation

The structured annotations, visualized through CATMA (distribution chart), Gephi (network graph), and Seaborn plots in Python (e.g., scatter plot) support thorough source critique and interpretation. Previously examined discourse elements can be quickly revisited and reinterpreted when new questions arise in light of additional newspaper articles. Moreover, diachronic comparisons are easily feasible.

This workflow seeks to methodologically advance research on the cultural history of politics of the Weimar Republic, specifically for examining discourse on democracy, which has not previously seen a systematic scalable reading methodology for large volumes of newspapers. It

enables more extensive, detailed, and nuanced insights into political discourse of Weimar's press. In doing so, the project also methodologically contributes to digital newspaper research at large – ensured by documenting and publishing all the code as Jupyter Notebooks, as well as the produced data and metadata on GitHub and the TextGrid Repository. This provides open documentation and reusability.

# Notes

1. The Python code, along with a prompt detailing the OCR-cleaning instructions, has been published in a Jupyter Notebook under an open license (Wachter and Jentsch 2025).

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